# Sound speed and resonance

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use travel time and resonant conditions to interpret sound.

**SUCCESS CRITERIA - I can...**

1. Account for round-trip distance in an echo.
2. Identify resonance as a response near a natural frequency.
3. Use an appropriate wavelength relationship.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

Two waves meet the same opening with wavelengths 3 cm and 5 cm. Which generally diffracts more? Explain the relevant comparison.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $d_{echo} = v\,\Delta t/2$; v = 340 m/s.
Echo time is a round trip. Resonance occurs near a natural mode under suitable damping.

**Worked reasoning:** In an ideal tube open at both ends, the fundamental wavelength is twice its length. For L=0.5 m and sound speed 340 m/s, f=340/1=340 Hz.

**Your turn:** Sketch the fundamental with displacement antinodes at both ends.

open-open tube

Draw a displacement pattern; label nodes and antinodes.

**Marzano check - circle:** ① ② ③

## 03 Read and interpret the model

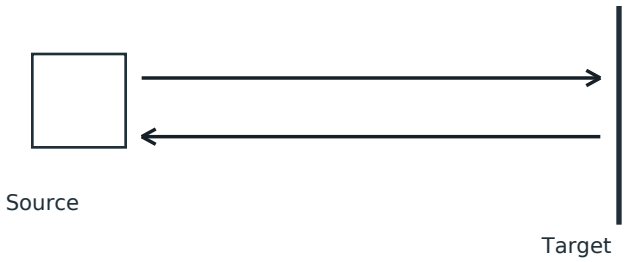
An echo returns from a wall. Draw the outgoing and returning path. Why does sound's total travel distance exceed wall range?




**Marzano check - circle:** ① ② ③

## 04 Apply the physics

In a scaled echo simulation, a pulse travels at 10 m/s and returns after 4 s. Label both legs of the trip. Calculate the total path distance and the one-way distance to the wall.




**Marzano check - circle:** ① ② ③

## 05 Explain your reasoning

Why does a resonant response not mean that any object can respond strongly to every frequency?

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A heat engine receives 600 J heat and outputs 180 J work per cycle. Find efficiency and rejected heat.
Label energy entering, leaving and stored.

Draw and label all energy transfers across the boundary.

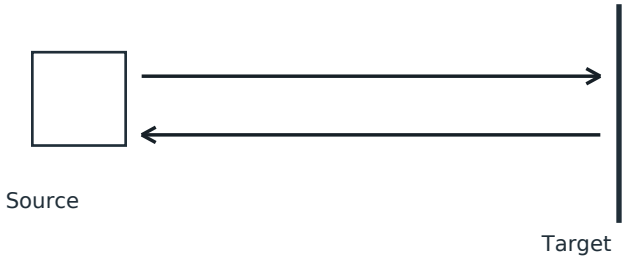
Chosen system

Stored energy change: _____ J

**Marzano check - circle:** 1 2 3

## 07 Exit check

An echo’s round-trip path is 60 m. How far away is the wall?




**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

Return to the worked example in problem 2. Compare fundamental frequencies for ideal open-open tubes of lengths 0.5 and 1.0 m at the same sound speed.

open-open tube

Draw a displacement pattern; label nodes and antinodes.

**Marzano check - circle:** 1 2 3